Benchmark results — Persistences, BLEND, AR-OLS, ELM cost-function variants, TimeGPT, and NeuralProphet

The runs use $N_{\text{hidden}} = 500$ and $N_{\text{draws}} = 100$, $\approx 35\,000$ samples (2 years, August 2020 – August 2022), without dimensionality reduction.
The Palaiseau plant has a peak power of 16.7 kWp (53 PV modules, $\approx 300\,\text{m}^2$ rooftop): the RMSE values reported below, in watts, should be read against this maximum (an RMSE of 1 000 W is $\approx 6\,\%$ of the installed capacity).
The ELM and AR-OLS inputs are the standardized window of the $LB = 48$ lags, augmented by 4 cyclical time features (sin/cos encoding of the hour of the day and of the day of the year) for the target time step.
Predictions are clipped to $[0, +\infty)$ because the models sometimes predicted negative values, which are physically impossible.
Most ELM variants select their hyperparameter(s) from a small grid by validation RMSE (20 % of the training set). The λ column reports the selected value (or the selected tuple for two-parameter variants, in the order stated in each variant's script). **Bold** marks the best NICE$^{\Sigma}$ among the cyclic ELM variants for each table (ties bolded together).

Two BLEND variants. **BLEND**$_{\text{optim}}$ (*optimized*): λ per phase learned by directly minimizing the MSE on the training set (closed-form OLS solution). **BLEND**$_{\text{corr}}$: λ derived from the phase-conditioned empirical correlation ($\lambda = (1+\rho)/2$), where ρ is the correlation between the current value x_t and the same phase of the previous day x_{t+h-T} (the P° term), conditioned on the issue-time phase $\phi(t)$. λ weights the cyclic term: $\hat{x}_{t+h} = \lambda\, x_{t+h-T} + (1-\lambda)\, x_t$. Both variants use the same persistences P and P°.

All data vs. daytime only. Each horizon is reported twice: *All data* includes all 30 min test steps (nights, where $P \approx 0$, dominate in count) and Daytime only, which restricts the evaluation to steps where the solar elevation at Palaiseau (48.71°N, 2.22°E) exceeds 0°.

External models. The bottom block of each table reports two external models, scored a posteriori with the same metrics. **The two are not directly comparable**: TimeGPT (Nixtla) is used *zero-shot*. The foundation model is never fitted on Palaiseau, it only receives a history window at inference, whereas NeuralProphet is *trained* on the first year of the series and predicts the second. TimeGPT is additionally evaluated on a subsampled test set (1 every 20 windows $\approx$875 windows), so its NICE is computed against persistences recomputed on that same subset. To ensure a fair comparison, the *NeuralProphet (clipped)* row clips the forecasts at 0,W. This is necessary because NeuralProphet's additive structure generates negative power values approximately 36% of the time at night, whereas the ELM handles this constraint internally by clipping its predictions. Once both models are bounded at zero, the comparison confirms that the tuned ELM still edges out NeuralProphet at every horizon.

FH = 0.5 h

Table 1: All data – $FH = 0.5\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE$^\Sigma$
Persistence P	0	–	–	–	1169.0	0.490	~ 0	0.239	0.910	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.793	1.966	1.938	1.899
BLEND$_{\mathrm{corr}}$	0	–	–	–	1850.1	0.775	+0.001	0.352	0.775	1.469	1.583	1.550	1.534
BLEND$_{\mathrm{optim}}$	48	–	–	–	1079.7	0.452	−0.014	0.216	0.923	0.901	0.924	0.931	0.918
AR-OLS	53	–	–	–	1012.4	0.424	+0.014	0.212	0.933	0.887	0.866	0.880	0.878
ELM (OLS)	27000	500	100	–	991.6	0.415	+0.021	0.212	0.935	0.884	0.848	0.861	0.865
ELM (Ridge)	27000	500	100	25	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (Rob. Risk)	27000	500	100	5	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (Tikhonov)	27000	500	100	25	987.9	0.414	+0.024	0.211	0.936	0.881	0.845	0.860	0.862
ELM (Box-Cox)	27000	500	100	(25,1)	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (MAE)	27000	500	100	(10,10)	983.7	0.412	+0.020	0.205	0.936	0.859	0.841	0.862	0.854
ELM (Log-MSE)	27000	500	100	(25,100)	1234.2	0.517	−0.043	0.238	0.900	0.993	1.056	1.073	1.041
ELM (Huber)	27000	500	100	25	983.3	0.412	+0.019	0.201	0.936	0.841	0.841	0.868	**0.850**
ELM (L3)	27000	500	100	(10,500)	1161.9	0.487	+0.050	0.288	0.911	1.202	0.994	0.936	1.044
ELM (Corr)	27000	500	100	(25,0.5)	997.6	0.418	+0.022	0.201	0.935	0.841	0.853	0.892	0.862
ELM (Elastic Net)	27000	500	100	(1,25)	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (M-Est.)	27000	500	100	25	985.0	0.413	+0.019	0.200	0.936	0.835	0.843	0.871	**0.850**
ELM (Lp)	27000	500	100	(25,1.5)	982.9	0.412	+0.021	0.205	0.937	0.855	0.841	0.863	0.853
ELM (GLM)	27000	500	100	10	1115.3	0.467	+0.085	0.274	0.918	1.144	0.954	0.940	1.013
ELM$_{\mathrm{no\,cyc}}$ (OLS)	25000	500	100	–	1000.2	0.419	+0.020	0.211	0.934	0.881	0.856	0.873	0.870
ELM$_{\mathrm{no\,cyc}}$ (Ridge)	25000	500	100	25	992.6	0.416	+0.016	0.208	0.935	0.869	0.849	0.866	0.861
TimeGPT	–	–	–	–	1115.5	0.472	−0.049	0.225	0.918	0.946	0.974	1.004	0.974
NeuralProphet	–	–	–	–	1005.9	0.420	−0.022	0.227	0.934	0.947	0.859	0.874	0.893
NeuralProphet (clipped)	–	–	–	–	996.4	0.416	+0.004	0.202	0.935	0.842	0.850	0.873	0.855
NeuralProphet$_{\mathrm{no\,seas.}}$	–	–	–	–	1314.1	0.549	−0.008	0.347	0.887	1.450	1.122	1.059	1.210

Table 2: Daytime only ($\theta_{\mathrm{sol}} > 0°$) – $FH = 0.5\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE$^\Sigma$
Persistence P	0	–	–	–	1650.0	0.347	~ 0	0.239	0.857	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.795	1.966	1.938	1.900
BLEND$_{\mathrm{corr}}$	0	–	–	–	2611.3	0.549	+0.001	0.352	0.642	1.470	1.583	1.550	1.534
BLEND$_{\mathrm{optim}}$	48	–	–	–	1524.0	0.320	−0.014	0.216	0.878	0.902	0.924	0.931	0.919
AR-OLS	53	–	–	–	1423.3	0.299	+0.001	0.200	0.894	0.836	0.863	0.880	0.859
ELM (OLS)	27000	500	100	–	1382.8	0.291	−0.001	0.189	0.900	0.793	0.838	0.860	0.830
ELM (Ridge)	27000	500	100	25	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (Rob. Risk)	27000	500	100	5	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (Tikhonov)	27000	500	100	25	1379.4	0.290	+0.002	0.189	0.900	0.791	0.836	0.859	0.829
ELM (Box-Cox)	27000	500	100	(25,1)	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (MAE)	27000	500	100	(10,10)	1376.6	0.289	+0.001	0.186	0.901	0.780	0.834	0.861	0.825
ELM (Log-MSE)	27000	500	100	(25,100)	1741.8	0.366	−0.046	0.235	0.841	0.983	1.056	1.073	1.037
ELM (Huber)	27000	500	100	25	1377.6	0.290	+0.002	0.184	0.900	0.768	0.835	0.868	**0.823**
ELM (L3)	27000	500	100	(10,500)	1576.5	0.331	+0.001	0.238	0.870	0.995	0.955	0.927	0.959
ELM (Corr)	27000	500	100	(25,0.5)	1394.6	0.293	+0.002	0.181	0.898	0.759	0.845	0.891	0.832
ELM (Elastic Net)	27000	500	100	(1,25)	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (M-Est.)	27000	500	100	25	1381.0	0.290	+0.002	0.183	0.900	0.766	0.837	0.871	0.825
ELM (Lp)	27000	500	100	(25,1.5)	1375.6	0.289	+0.002	0.186	0.901	0.777	0.834	0.862	0.824
ELM (GLM)	27000	500	100	10	1532.9	0.322	+0.035	0.224	0.877	0.936	0.929	0.936	0.934
ELM$_{\mathrm{no\,cyc}}$ (OLS)	25000	500	100	–	1397.1	0.294	−0.001	0.190	0.898	0.796	0.847	0.872	0.838
ELM$_{\mathrm{no\,cyc}}$ (Ridge)	25000	500	100	25	1390.8	0.292	−0.002	0.190	0.899	0.797	0.843	0.866	0.835
TimeGPT	–	–	–	–	1573.8	0.334	−0.053	0.220	0.871	0.928	0.972	1.004	0.968
NeuralProphet	–	–	–	–	1404.4	0.295	−0.006	0.196	0.897	0.822	0.850	0.873	0.848
NeuralProphet (clipped)	–	–	–	–	1402.5	0.294	−0.004	0.194	0.897	0.812	0.848	0.873	0.844
NeuralProphet$_{\mathrm{no\,seas.}}$	–	–	–	–	1772.5	0.372	+0.005	0.264	0.835	1.104	1.072	1.049	1.075

FH = 1 h

Table 3: All data – $FH = 1\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE^Σ
Persistence P	0	–	–	–	1699.5	0.712	~ 0	0.379	0.810	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.133	1.352	1.429	1.305
$\text{BLEND}_{\text{corr}}$	0	–	–	–	1885.6	0.790	+0.001	0.364	0.766	0.962	1.110	1.163	1.078
$\text{BLEND}_{\text{optim}}$	48	–	–	–	1379.6	0.578	−0.031	0.283	0.875	0.746	0.812	0.848	0.802
AR-OLS	53	–	–	–	1277.0	0.535	+0.022	0.285	0.893	0.753	0.751	0.785	0.763
ELM (OLS)	27000	500	100	–	1207.0	0.506	+0.024	0.269	0.904	0.709	0.710	0.751	0.724
ELM (Ridge)	27000	500	100	25	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (Rob. Risk)	27000	500	100	5	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (Tikhonov)	27000	500	100	25	1203.0	0.504	+0.021	0.267	0.905	0.703	0.708	0.749	0.720
ELM (Box-Cox)	27000	500	100	(10,0.75)	1197.6	0.502	−0.019	0.246	0.906	0.648	0.705	0.756	**0.703**
ELM (MAE)	27000	500	100	(25,10)	1200.9	0.503	+0.015	0.259	0.905	0.683	0.707	0.753	0.714
ELM (Log-MSE)	27000	500	100	(25,100)	1405.4	0.589	−0.073	0.275	0.870	0.726	0.827	0.888	0.814
ELM (Huber)	27000	500	100	25	1203.1	0.504	+0.016	0.255	0.905	0.672	0.708	0.760	0.713
ELM (L3)	27000	500	100	(10,500)	1380.7	0.578	+0.053	0.356	0.875	0.939	0.812	0.796	0.849
ELM (Corr)	27000	500	100	(25,0.5)	1213.1	0.508	+0.027	0.262	0.903	0.692	0.714	0.771	0.726
ELM (Elastic Net)	27000	500	100	(10,25)	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (M-Est.)	27000	500	100	10	1214.2	0.509	+0.015	0.253	0.903	0.669	0.714	0.770	0.718
ELM (Lp)	27000	500	100	(25,1.25)	1199.7	0.502	+0.016	0.258	0.905	0.682	0.706	0.753	0.714
ELM (GLM)	27000	500	100	10	1288.5	0.540	+0.062	0.303	0.891	0.798	0.758	0.797	0.784
$\text{ELM}_{\text{no cyc}}$ (OLS)	25000	500	100	–	1223.8	0.513	+0.021	0.270	0.902	0.712	0.720	0.763	0.732
$\text{ELM}_{\text{no cyc}}$ (Ridge)	25000	500	100	25	1218.1	0.510	+0.016	0.266	0.903	0.702	0.717	0.762	0.727
TimeGPT	–	–	–	–	1650.2	0.694	−0.033	0.398	0.819	1.037	0.957	0.968	0.988
NeuralProphet	–	–	–	–	1262.6	0.527	−0.036	0.309	0.896	0.815	0.741	0.771	0.776
NeuralProphet (clipped)	–	–	–	–	1241.9	0.519	+0.006	0.267	0.899	0.705	0.729	0.769	0.734
$\text{NeuralProphet}_{\text{no seas.}}$	–	–	–	–	1478.9	0.618	−0.002	0.408	0.857	1.076	0.868	0.849	0.931

Table 4: Daytime only ($\theta_{\text{sol}} > 0°$) – $FH = 1\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE^Σ
Persistence P	0	–	–	–	2397.6	0.504	−0.003	0.376	0.698	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.140	1.353	1.429	1.308
$\text{BLEND}_{\text{corr}}$	0	–	–	–	2661.5	0.560	~ 0	0.364	0.628	0.967	1.110	1.163	1.080
$\text{BLEND}_{\text{optim}}$	48	–	–	–	1947.2	0.409	−0.031	0.282	0.801	0.750	0.812	0.848	0.804
AR-OLS	53	–	–	–	1792.6	0.377	+0.004	0.267	0.831	0.709	0.748	0.784	0.747
ELM (OLS)	27000	500	100	–	1682.6	0.354	−0.004	0.241	0.851	0.639	0.702	0.750	0.697
ELM (Ridge)	27000	500	100	25	1681.2	0.353	−0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (Rob. Risk)	27000	500	100	5	1681.2	0.353	−0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (Tikhonov)	27000	500	100	25	1680.3	0.353	−0.005	0.241	0.852	0.639	0.701	0.748	0.696
ELM (Box-Cox)	27000	500	100	(10,0.75)	1687.5	0.355	−0.029	0.236	0.851	0.627	0.704	0.756	0.696
ELM (MAE)	27000	500	100	(25,10)	1683.2	0.354	−0.005	0.238	0.851	0.632	0.702	0.753	0.696
ELM (Log-MSE)	27000	500	100	(25,100)	1983.4	0.417	−0.075	0.273	0.794	0.724	0.827	0.888	0.813
ELM (Huber)	27000	500	100	25	1686.6	0.355	−0.005	0.234	0.851	0.622	0.703	0.760	**0.695**
ELM (L3)	27000	500	100	(10,500)	1877.3	0.395	−0.005	0.298	0.815	0.791	0.783	0.789	0.787
ELM (Corr)	27000	500	100	(25,0.5)	1692.9	0.356	~ 0	0.235	0.850	0.624	0.706	0.770	0.700
ELM (Elastic Net)	27000	500	100	(10,25)	1681.2	0.353	−0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (M-Est.)	27000	500	100	10	1704.6	0.358	−0.003	0.235	0.848	0.625	0.711	0.770	0.702
ELM (Lp)	27000	500	100	(25,1.25)	1681.1	0.353	−0.005	0.237	0.852	0.630	0.701	0.753	**0.695**
ELM (GLM)	27000	500	100	10	1796.5	0.378	+0.023	0.264	0.831	0.702	0.749	0.796	0.749
$\text{ELM}_{\text{no cyc}}$ (OLS)	25000	500	100	–	1709.8	0.360	−0.005	0.244	0.847	0.649	0.713	0.762	0.708
$\text{ELM}_{\text{no cyc}}$ (Ridge)	25000	500	100	25	1707.6	0.359	−0.006	0.244	0.847	0.649	0.712	0.762	0.708
TimeGPT	–	–	–	–	2275.1	0.482	−0.076	0.355	0.725	0.931	0.937	0.963	0.944
NeuralProphet	–	–	–	–	1750.9	0.367	−0.008	0.260	0.839	0.692	0.729	0.769	0.730
NeuralProphet (clipped)	–	–	–	–	1747.1	0.367	−0.004	0.257	0.840	0.683	0.727	0.768	0.726
$\text{NeuralProphet}_{\text{no seas.}}$	–	–	–	–	1955.5	0.410	+0.012	0.298	0.800	0.791	0.814	0.834	0.813

FH = 3 h

Table 5: All data – $FH = 3\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE[1]	NICE[2]	NICE[3]	NICE[Σ]
Persistence P	0	–	–	–	3594.0	1.505	-0.001	0.890	0.151	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	2298.5	0.963	~ 0	0.429	0.653	0.483	0.640	0.727	0.616
$\mathrm{BLEND_{corr}}$	0	–	–	–	2118.7	0.887	-0.002	0.473	0.705	0.532	0.590	0.645	0.589
$\mathrm{BLEND_{optim}}$	48	–	–	–	1914.3	0.802	-0.072	0.383	0.759	0.430	0.533	0.597	0.520
AR-OLS	53	–	–	–	1800.2	0.754	$+0.029$	0.425	0.787	0.478	0.501	0.540	0.506
ELM (OLS)	27000	500	100	–	1665.3	0.697	$+0.026$	0.385	0.818	0.433	0.463	0.506	0.468
ELM (Ridge)	27000	500	100	25	1660.3	0.695	$+0.017$	0.382	0.819	0.429	0.462	0.505	0.465
ELM (Rob. Risk)	27000	500	100	5	1660.3	0.695	$+0.017$	0.382	0.819	0.429	0.462	0.505	0.465
ELM (Tikhonov)	27000	500	100	25	1660.6	0.695	$+0.021$	0.383	0.819	0.430	0.462	0.505	0.466
ELM (Box-Cox)	27000	500	100	(25,0.75)	1663.4	0.697	-0.043	0.357	0.818	0.401	0.463	0.513	**0.459**
ELM (MAE)	27000	500	100	(25,100)	1673.6	0.701	$+0.005$	0.368	0.816	0.413	0.466	0.517	0.465
ELM (Log-MSE)	27000	500	100	(25,100)	1887.9	0.791	-0.145	0.378	0.766	0.424	0.525	0.590	0.513
ELM (Huber)	27000	500	100	25	1679.3	0.703	$+0.005$	0.364	0.815	0.409	0.467	0.521	0.466
ELM (L3)	27000	500	100	(10,500)	1822.3	0.763	$+0.081$	0.483	0.782	0.542	0.507	0.518	0.522
ELM (Corr)	27000	500	100	(25,0.5)	1660.9	0.696	$+0.032$	0.384	0.819	0.432	0.462	0.506	0.467
ELM (Elastic Net)	27000	500	100	(1,25)	1660.3	0.695	$+0.017$	0.382	0.819	0.429	0.462	0.505	0.465
ELM (M-Est.)	27000	500	100	25	1699.6	0.712	-0.003	0.360	0.810	0.404	0.473	0.531	0.470
ELM (Lp)	27000	500	100	(25,1.25)	1670.6	0.700	$+0.010$	0.370	0.817	0.416	0.465	0.515	0.465
ELM (GLM)	27000	500	100	10	1737.0	0.727	$+0.056$	0.405	0.802	0.455	0.483	0.535	0.491
$\mathrm{ELM_{no\,cyc}}$ (OLS)	25000	500	100	–	1713.1	0.717	$+0.020$	0.392	0.807	0.441	0.477	0.523	0.480
$\mathrm{ELM_{no\,cyc}}$ (Ridge)	25000	500	100	25	1705.2	0.714	$+0.012$	0.388	0.809	0.436	0.474	0.521	0.477
TimeGPT	–	–	–	–	3412.1	1.436	$+0.041$	0.977	0.223	1.093	0.950	0.924	0.989
NeuralProphet	–	–	–	–	1775.0	0.741	-0.081	0.467	0.794	0.526	0.493	0.526	0.515
NeuralProphet (clipped)	–	–	–	–	1734.0	0.724	-0.003	0.389	0.803	0.438	0.482	0.524	0.481
$\mathrm{NeuralProphet_{no\,seas.}}$	–	–	–	–	2054.6	0.858	-0.006	0.553	0.723	0.622	0.571	0.607	0.600

Table 6: Daytime only ($\theta_{\mathrm{sol}} > 0°$) – $FH = 3\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE[1]	NICE[2]	NICE[3]	NICE[Σ]
Persistence P	0	–	–	–	4919.5	1.034	-0.082	0.808	-0.270	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	3244.4	0.682	~ 0	0.429	0.448	0.531	0.660	0.738	0.643
$\mathrm{BLEND_{corr}}$	0	–	–	–	2965.9	0.623	-0.029	0.446	0.538	0.552	0.603	0.654	0.603
$\mathrm{BLEND_{optim}}$	48	–	–	–	2701.8	0.568	-0.073	0.382	0.617	0.473	0.549	0.606	0.543
AR-OLS	53	–	–	–	2520.8	0.530	-0.001	0.395	0.667	0.489	0.512	0.548	0.516
ELM (OLS)	27000	500	100	–	2326.0	0.489	-0.010	0.349	0.716	0.433	0.473	0.513	0.473
ELM (Ridge)	27000	500	100	25	2325.3	0.489	-0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (Rob. Risk)	27000	500	100	5	2325.3	0.489	-0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (Tikhonov)	27000	500	100	25	2323.3	0.488	-0.012	0.350	0.717	0.433	0.472	0.512	0.472
ELM (Box-Cox)	27000	500	100	(25,0.75)	2343.8	0.493	-0.055	0.344	0.712	0.426	0.476	0.520	0.474
ELM (MAE)	27000	500	100	(25,100)	2350.0	0.494	-0.020	0.342	0.710	0.424	0.478	0.524	0.475
ELM (Log-MSE)	27000	500	100	(25,100)	2664.7	0.560	-0.149	0.374	0.627	0.463	0.542	0.598	0.534
ELM (Huber)	27000	500	100	25	2357.8	0.496	-0.020	0.338	0.708	0.419	0.479	0.529	0.476
ELM (L3)	27000	500	100	(10,500)	2487.8	0.523	$+0.008$	0.410	0.675	0.507	0.506	0.521	0.511
ELM (Corr)	27000	500	100	(25,0.5)	2316.0	0.487	-0.006	0.346	0.719	0.428	0.471	0.512	**0.471**
ELM (Elastic Net)	27000	500	100	(1,25)	2325.3	0.489	-0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (M-Est.)	27000	500	100	25	2388.8	0.502	-0.026	0.337	0.701	0.417	0.486	0.539	0.481
ELM (Lp)	27000	500	100	(25,1.25)	2344.2	0.493	-0.017	0.343	0.712	0.425	0.477	0.522	0.475
ELM (GLM)	27000	500	100	10	2433.9	0.512	$+0.014$	0.363	0.689	0.449	0.495	0.543	0.496
$\mathrm{ELM_{no\,cyc}}$ (OLS)	25000	500	100	–	2396.0	0.504	-0.016	0.356	0.699	0.441	0.487	0.530	0.486
$\mathrm{ELM_{no\,cyc}}$ (Ridge)	25000	500	100	25	2390.1	0.502	-0.019	0.356	0.700	0.441	0.486	0.529	0.485
TimeGPT	–	–	–	–	4304.5	0.910	-0.229	0.706	0.009	0.859	0.870	0.884	0.871
NeuralProphet	–	–	–	–	2441.6	0.512	-0.023	0.377	0.688	0.467	0.496	0.532	0.498
NeuralProphet (clipped)	–	–	–	–	2437.7	0.512	-0.019	0.373	0.689	0.462	0.495	0.532	0.496
$\mathrm{NeuralProphet_{no\,seas.}}$	–	–	–	–	2783.2	0.584	$+0.018$	0.425	0.594	0.526	0.565	0.611	0.567

FH = 6 h

Table 7: All data – $FH = 6\,\text{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE^Σ
Persistence P	0	–	–	–	5509.8	2.307	~ 0	1.510	-0.994	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	2298.7	0.963	~ 0	0.429	0.653	0.284	0.417	0.494	0.398
$\text{BLEND}_{\text{corr}}$	0	–	–	–	2819.8	1.181	−0.001	0.744	0.478	0.493	0.512	0.536	0.513
$\text{BLEND}_{\text{optim}}$	48	–	–	–	2164.7	0.906	−0.115	0.423	0.692	0.280	0.393	0.463	0.379
AR-OLS	53	–	–	–	1960.5	0.821	+0.017	0.458	0.748	0.303	0.356	0.402	0.354
ELM (OLS)	27000	500	100	–	1879.7	0.787	+0.014	0.436	0.768	0.289	0.341	0.385	0.338
ELM (Ridge)	27000	500	100	25	1859.9	0.779	+0.007	0.425	0.773	0.282	0.338	0.382	0.334
ELM (Rob. Risk)	27000	500	100	10	1863.4	0.780	+0.001	0.425	0.772	0.281	0.338	0.383	0.334
ELM (Tikhonov)	27000	500	100	25	1871.9	0.784	+0.010	0.432	0.770	0.286	0.340	0.384	0.336
ELM (Box-Cox)	27000	500	100	(25,0.75)	1870.5	0.783	−0.055	0.405	0.770	0.268	0.339	0.387	**0.332**
ELM (MAE)	27000	500	100	(25,100)	1883.5	0.789	−0.005	0.413	0.767	0.274	0.342	0.393	0.336
ELM (Log-MSE)	27000	500	100	(25,100)	2057.9	0.862	−0.179	0.423	0.722	0.280	0.374	0.428	0.360
ELM (Huber)	27000	500	100	25	1895.0	0.793	−0.005	0.410	0.764	0.272	0.344	0.397	0.338
ELM (L3)	27000	500	100	(10,500)	2027.2	0.849	+0.064	0.530	0.730	0.351	0.368	0.389	0.369
ELM (Corr)	27000	500	100	(25,0.5)	1886.4	0.790	+0.020	0.439	0.766	0.291	0.342	0.386	0.340
ELM (Elastic Net)	27000	500	100	(10,25)	1859.9	0.779	+0.007	0.425	0.773	0.282	0.338	0.382	0.334
ELM (M-Est.)	27000	500	100	25	1916.2	0.802	−0.011	0.406	0.759	0.269	0.348	0.405	0.340
ELM (Lp)	27000	500	100	(25,1.25)	1878.4	0.787	~ 0	0.416	0.768	0.275	0.341	0.391	0.336
ELM (GLM)	27000	500	100	10	1935.2	0.810	+0.060	0.458	0.754	0.303	0.351	0.398	0.351
$\text{ELM}_{\text{no cyc}}$ (OLS)	25000	500	100	–	1930.9	0.809	+0.007	0.448	0.755	0.297	0.350	0.398	0.348
$\text{ELM}_{\text{no cyc}}$ (Ridge)	25000	500	100	25	1909.3	0.799	−0.002	0.439	0.761	0.291	0.347	0.393	0.343
TimeGPT	–	–	–	–	5116.8	2.074	+0.179	1.561	-0.646	1.078	0.929	0.896	0.968
NeuralProphet	–	–	–	–	1959.6	0.818	−0.114	0.505	0.748	0.334	0.355	0.399	0.363
NeuralProphet (clipped)	–	–	–	–	1919.9	0.802	−0.027	0.418	0.759	0.277	0.348	0.398	0.341
$\text{NeuralProphet}_{\text{no seas.}}$	–	–	–	–	2190.6	0.915	+0.003	0.554	0.686	0.367	0.397	0.451	0.405

Table 8: Daytime only ($\theta_{\text{sol}} > 0°$) – $FH = 6\,\text{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE^Σ
Persistence P	0	–	–	–	6756.5	1.420	−0.408	1.103	-1.395	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.389	0.480	0.539	0.469
$\text{BLEND}_{\text{corr}}$	0	–	–	–	3596.5	0.756	−0.181	0.564	0.321	0.511	0.532	0.557	0.534
$\text{BLEND}_{\text{optim}}$	48	–	–	–	3055.4	0.642	−0.115	0.423	0.510	0.383	0.452	0.505	0.447
AR-OLS	53	–	–	–	2750.6	0.578	−0.014	0.427	0.603	0.387	0.407	0.438	0.411
ELM (OLS)	27000	500	100	–	2631.2	0.553	−0.022	0.400	0.637	0.362	0.389	0.419	0.390
ELM (Ridge)	27000	500	100	25	2610.1	0.549	−0.023	0.396	0.643	0.359	0.386	0.417	**0.387**
ELM (Rob. Risk)	27000	500	100	10	2618.0	0.550	−0.025	0.399	0.640	0.362	0.387	0.418	0.389
ELM (Tikhonov)	27000	500	100	25	2624.1	0.552	−0.023	0.398	0.639	0.361	0.388	0.418	0.389
ELM (Box-Cox)	27000	500	100	(25,0.75)	2636.5	0.554	−0.068	0.392	0.635	0.356	0.390	0.422	0.389
ELM (MAE)	27000	500	100	(25,100)	2648.4	0.557	−0.029	0.389	0.632	0.352	0.392	0.428	0.391
ELM (Log-MSE)	27000	500	100	(25,100)	2904.6	0.611	−0.182	0.419	0.557	0.380	0.430	0.467	0.426
ELM (Huber)	27000	500	100	25	2664.0	0.560	−0.030	0.385	0.628	0.349	0.394	0.433	0.392
ELM (L3)	27000	500	100	(10,500)	2782.0	0.585	−0.006	0.460	0.594	0.417	0.412	0.420	0.416
ELM (Corr)	27000	500	100	(25,0.5)	2635.9	0.554	−0.020	0.400	0.635	0.362	0.390	0.420	0.391
ELM (Elastic Net)	27000	500	100	(10,25)	2610.1	0.549	−0.023	0.396	0.643	0.359	0.386	0.417	**0.387**
ELM (M-Est.)	27000	500	100	25	2696.0	0.567	−0.034	0.383	0.619	0.347	0.399	0.441	0.396
ELM (Lp)	27000	500	100	(25,1.25)	2639.8	0.555	−0.026	0.390	0.634	0.353	0.391	0.426	0.390
ELM (GLM)	27000	500	100	10	2712.4	0.570	+0.014	0.413	0.614	0.374	0.401	0.433	0.403
$\text{ELM}_{\text{no cyc}}$ (OLS)	25000	500	100	–	2704.2	0.568	−0.031	0.409	0.616	0.371	0.400	0.433	0.402
$\text{ELM}_{\text{no cyc}}$ (Ridge)	25000	500	100	25	2679.8	0.563	−0.035	0.406	0.623	0.368	0.397	0.429	0.398
TimeGPT	–	–	–	–	5687.2	1.155	−0.511	0.872	-0.645	0.823	0.837	0.852	0.837
NeuralProphet	–	–	–	–	2702.9	0.567	−0.043	0.404	0.617	0.367	0.400	0.434	0.400
NeuralProphet (clipped)	–	–	–	–	2702.4	0.567	−0.042	0.403	0.617	0.366	0.400	0.434	0.400
$\text{NeuralProphet}_{\text{no seas.}}$	–	–	–	–	3000.2	0.630	+0.004	0.437	0.528	0.396	0.444	0.489	0.443

$\mathbf{FH = 10\,h}$

Table 9: All data – $FH = 10\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE^Σ
Persistence P	0	–	–	–	6403.1	2.680	~ 0	1.927	-1.693	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	2298.9	0.962	~ 0	0.429	0.653	0.223	0.359	0.440	0.341
$\mathrm{BLEND_{corr}}$	0	–	–	–	3488.1	1.460	−0.002	1.005	0.201	0.521	0.545	0.565	0.544
$\mathrm{BLEND_{optim}}$	48	–	–	–	2224.0	0.931	−0.134	0.433	0.675	0.225	0.347	0.423	0.332
AR-OLS	53	–	–	–	1964.9	0.823	+0.014	0.458	0.746	0.238	0.307	0.359	0.301
ELM (OLS)	27000	500	100	–	1894.3	0.793	+0.016	0.439	0.764	0.228	0.296	0.345	0.289
ELM (Ridge)	27000	500	100	25	1887.3	0.790	+0.008	0.434	0.766	0.225	0.295	0.344	0.288
ELM (Rob. Risk)	27000	500	100	10	1888.8	0.791	−0.003	0.432	0.766	0.224	0.295	0.344	0.288
ELM (Tikhonov)	27000	500	100	25	1889.3	0.791	+0.011	0.436	0.766	0.226	0.295	0.344	0.288
ELM (Box-Cox)	27000	500	100	(25,0.75)	1899.9	0.795	−0.061	0.412	0.763	0.214	0.297	0.349	**0.286**
ELM (MAE)	27000	500	100	(25,100)	1903.5	0.797	−0.011	0.417	0.762	0.217	0.297	0.352	0.289
ELM (Log-MSE)	27000	500	100	(25,100)	2087.2	0.874	−0.189	0.433	0.714	0.225	0.326	0.383	0.311
ELM (Huber)	27000	500	100	25	1914.6	0.801	−0.011	0.414	0.759	0.215	0.299	0.356	0.290
ELM (L3)	27000	500	100	(10,500)	2078.3	0.870	+0.062	0.549	0.716	0.285	0.325	0.353	0.321
ELM (Corr)	27000	500	100	(25,0.5)	1900.7	0.796	+0.014	0.440	0.763	0.228	0.297	0.345	0.290
ELM (Elastic Net)	27000	500	100	(10,25)	1887.3	0.790	+0.008	0.434	0.766	0.225	0.295	0.344	0.288
ELM (M-Est.)	27000	500	100	25	1931.0	0.808	−0.017	0.410	0.755	0.213	0.302	0.361	0.292
ELM (Lp)	27000	500	100	(25,1.25)	1900.7	0.796	−0.006	0.420	0.763	0.218	0.297	0.351	0.289
ELM (GLM)	27000	500	100	10	1972.5	0.826	+0.051	0.458	0.744	0.238	0.308	0.362	0.302
$\mathrm{ELM_{no\,cyc}}$ (OLS)	25000	500	100	–	1948.3	0.816	+0.004	0.451	0.751	0.234	0.304	0.355	0.298
$\mathrm{ELM_{no\,cyc}}$ (Ridge)	25000	500	100	25	1940.2	0.812	−0.003	0.444	0.753	0.231	0.303	0.354	0.296
TimeGPT	–	–	–	–	5131.1	2.206	+0.414	1.721	-0.805	0.901	0.821	0.799	0.840
NeuralProphet	–	–	–	–	1974.8	0.825	−0.132	0.513	0.745	0.266	0.308	0.357	0.311
NeuralProphet (clipped)	–	–	–	–	1931.1	0.807	−0.038	0.419	0.756	0.218	0.301	0.356	0.292
$\mathrm{NeuralProphet_{no\,seas.}}$	–	–	–	–	2175.0	0.908	−0.004	0.554	0.690	0.288	0.339	0.398	0.342

Table 10: Daytime only ($\theta_{\mathrm{sol}} > 0°$) – $FH = 10\,\mathrm{h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE^1	NICE^2	NICE^3	NICE^Σ
Persistence P	0	–	–	–	6688.9	1.406	−0.852	1.074	-1.348	1.000	1.000	1.000	1.000
Persistence P°	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.399	0.485	0.540	0.475
$\mathrm{BLEND_{corr}}$	0	–	–	–	3917.5	0.823	−0.415	0.592	0.195	0.551	0.586	0.614	0.583
$\mathrm{BLEND_{optim}}$	48	–	–	–	3138.9	0.660	−0.135	0.433	0.483	0.403	0.469	0.519	0.464
AR-OLS	53	–	–	–	2758.3	0.580	−0.016	0.429	0.601	0.399	0.412	0.440	0.417
ELM (OLS)	27000	500	100	–	2653.2	0.558	−0.019	0.404	0.631	0.376	0.397	0.423	**0.398**
ELM (Ridge)	27000	500	100	25	2648.3	0.557	−0.022	0.404	0.632	0.376	0.396	0.422	**0.398**
ELM (Rob. Risk)	27000	500	100	10	2655.3	0.558	−0.027	0.407	0.630	0.379	0.397	0.422	0.399
ELM (Tikhonov)	27000	500	100	25	2649.2	0.557	−0.021	0.404	0.632	0.376	0.396	0.422	**0.398**
ELM (Box-Cox)	27000	500	100	(25,0.75)	2678.6	0.563	−0.072	0.401	0.624	0.373	0.400	0.428	0.401
ELM (MAE)	27000	500	100	(25,100)	2678.7	0.563	−0.033	0.396	0.624	0.368	0.400	0.432	0.400
ELM (Log-MSE)	27000	500	100	(25,100)	2945.6	0.619	−0.192	0.430	0.545	0.400	0.440	0.471	0.437
ELM (Huber)	27000	500	100	25	2694.0	0.566	−0.033	0.392	0.619	0.365	0.403	0.437	0.402
ELM (L3)	27000	500	100	(10,500)	2845.0	0.598	−0.013	0.474	0.575	0.442	0.425	0.429	0.432
ELM (Corr)	27000	500	100	(25,0.5)	2662.7	0.560	−0.020	0.406	0.628	0.378	0.398	0.424	0.400
ELM (Elastic Net)	27000	500	100	(10,25)	2648.3	0.557	−0.022	0.404	0.632	0.376	0.396	0.422	**0.398**
ELM (M-Est.)	27000	500	100	25	2718.6	0.571	−0.037	0.390	0.612	0.363	0.406	0.444	0.404
ELM (Lp)	27000	500	100	(25,1.25)	2673.5	0.562	−0.029	0.397	0.625	0.369	0.400	0.431	0.400
ELM (GLM)	27000	500	100	10	2769.9	0.582	+0.015	0.422	0.597	0.393	0.414	0.444	0.417
$\mathrm{ELM_{no\,cyc}}$ (OLS)	25000	500	100	–	2734.2	0.575	−0.029	0.417	0.608	0.388	0.409	0.436	0.411
$\mathrm{ELM_{no\,cyc}}$ (Ridge)	25000	500	100	25	2726.5	0.573	−0.032	0.416	0.610	0.387	0.408	0.435	0.410
TimeGPT	–	–	–	–	5053.0	1.088	−0.492	0.816	-0.390	0.770	0.773	0.783	0.775
NeuralProphet	–	–	–	–	2719.8	0.571	−0.052	0.407	0.612	0.379	0.406	0.437	0.407
NeuralProphet (clipped)	–	–	–	–	2719.4	0.571	−0.051	0.406	0.613	0.378	0.406	0.437	0.407
$\mathrm{NeuralProphet_{no\,seas.}}$	–	–	–	–	2986.5	0.627	+0.007	0.441	0.533	0.410	0.446	0.487	0.448